Long-Run Economic Growth: Sources and Policies

ECON 101 – Principles of Macroeconomics

Muhebullah Karimzada

Spring 2026

Chapter Outline

- **7.1** The Importance of Economic Growth
- **7.2** The Economic Growth Model
- **7.3** New Growth Theory and Knowledge Capital
- **7.4** Why Isn't the Whole World Rich?

Chapter Outline

- **7.1** The Importance of Economic Growth
- **7.2** The Economic Growth Model
- **7.3** New Growth Theory and Knowledge Capital
- **7.4** Why Isn't the Whole World Rich?

Chapter Outline

- **7.1** The Importance of Economic Growth
- **7.2** The Economic Growth Model
- **7.3** New Growth Theory and Knowledge Capital
- **7.4** Why Isn't the Whole World Rich?

7.1 Learning Objective

Goal

Understand why economic growth matters and how growth rates differ across countries.

- Previous chapter: how to **measure** growth (real GDP, real GDP per capita).
- This chapter: how to **explain** and **influence** long-run growth.
- Key idea: economic growth is not automatic.
- History includes long periods of stagnation with little increase in output per person.

7.1 Learning Objective

Goal

Understand why economic growth matters and how growth rates differ across countries.

- Previous chapter: how to **measure** growth (real GDP, real GDP per capita).
- This chapter: how to **explain** and **influence** long-run growth.
- Key idea: economic growth is not automatic.
- History includes long periods of stagnation with little increase in output per person.

7.1 Learning Objective

Goal

Understand why economic growth matters and how growth rates differ across countries.

- Previous chapter: how to **measure** growth (real GDP, real GDP per capita).
- This chapter: how to **explain** and **influence** long-run growth.
- Key idea: economic growth is not automatic.
- History includes long periods of stagnation with little increase in output per person.

7.1 Learning Objective

Goal

Understand why economic growth matters and how growth rates differ across countries.

- Previous chapter: how to **measure** growth (real GDP, real GDP per capita).
- This chapter: how to **explain** and **influence** long-run growth.
- Key idea: economic growth is not automatic.
- History includes long periods of stagnation with little increase in output per person.

The Industrial Revolution

- **Industrial Revolution:** application of mechanical power to production.
  - Began in England around 1750.
  - Spread to North America by mid-1800s.
- Before that: real GDP per capita was essentially **flat** for centuries.
- Mechanical power enabled sustained long-run economic growth.
- **Canada's economic takeoff:**
  - Wheat exports, railway construction (CPR completed 1885), and industrialization drove rapid growth 1870–1930.
  - Resource extraction (forestry, mining, oil) and manufacturing (auto sector) fuelled 20th-century prosperity.

The Industrial Revolution

- **Industrial Revolution:** application of mechanical power to production.
  - Began in England around 1750.
  - Spread to North America by mid-1800s.
- Before that: real GDP per capita was essentially **flat** for centuries.
- Mechanical power enabled sustained long-run economic growth.
- **Canada's economic takeoff:**
  - Wheat exports, railway construction (CPR completed 1885), and industrialization drove rapid growth 1870–1930.
  - Resource extraction (forestry, mining, oil) and manufacturing (auto sector) fuelled 20th-century prosperity.

The Industrial Revolution

- **Industrial Revolution:** application of mechanical power to production.
  - Began in England around 1750.
  - Spread to North America by mid-1800s.
- Before that: real GDP per capita was essentially **flat** for centuries.
- Mechanical power enabled sustained long-run economic growth.
- **Canada's economic takeoff:**
  - Wheat exports, railway construction (CPR completed 1885), and industrialization drove rapid growth 1870–1930.
  - Resource extraction (forestry, mining, oil) and manufacturing (auto sector) fuelled 20th-century prosperity.

The Industrial Revolution

- **Industrial Revolution:** application of mechanical power to production.
  - Began in England around 1750.
  - Spread to North America by mid-1800s.
- Before that: real GDP per capita was essentially **flat** for centuries.
- Mechanical power enabled sustained long-run economic growth.
- **Canada's economic takeoff:**
  - Wheat exports, railway construction (CPR completed 1885), and industrialization drove rapid growth 1870–1930.
  - Resource extraction (forestry, mining, oil) and manufacturing (auto sector) fuelled 20th-century prosperity.

The Industrial Revolution

- **Industrial Revolution:** application of mechanical power to production.
  - Began in England around 1750.
  - Spread to North America by mid-1800s.
- Before that: real GDP per capita was essentially **flat** for centuries.
- Mechanical power enabled sustained long-run economic growth.
- Canada's economic takeoff:
  - Wheat exports, railway construction (CPR completed 1885), and industrialization drove rapid growth 1870–1930.
  - Resource extraction (forestry, mining, oil) and manufacturing (auto sector) fuelled 20th-century prosperity.

The Industrial Revolution

- **Industrial Revolution:** application of mechanical power to production.
  - Began in England around 1750.
  - Spread to North America by mid-1800s.
- Before that: real GDP per capita was essentially **flat** for centuries.
- Mechanical power enabled sustained long-run economic growth.
- **Canada's economic takeoff:**
  - Wheat exports, railway construction (CPR completed 1885), and industrialization drove rapid growth 1870–1930.
  - Resource extraction (forestry, mining, oil) and manufacturing (auto sector) fuelled 20th-century prosperity.

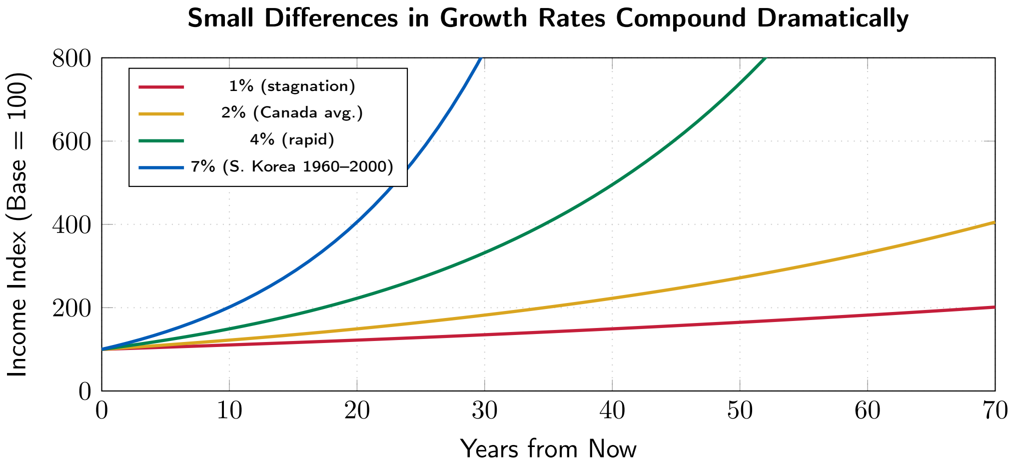
The Industrial Revolution

- **Industrial Revolution:** application of mechanical power to production.
  - Began in England around 1750.
  - Spread to North America by mid-1800s.
- Before that: real GDP per capita was essentially **flat** for centuries.
- Mechanical power enabled sustained long-run economic growth.
- **Canada's economic takeoff:**
  - Wheat exports, railway construction (CPR completed 1885), and industrialization drove rapid growth 1870–1930.
  - Resource extraction (forestry, mining, oil) and manufacturing (auto sector) fuelled 20th-century prosperity.

The Industrial Revolution

- **Industrial Revolution:** application of mechanical power to production.
  - Began in England around 1750.
  - Spread to North America by mid-1800s.
- Before that: real GDP per capita was essentially **flat** for centuries.
- Mechanical power enabled sustained long-run economic growth.
- **Canada's economic takeoff:**
  - Wheat exports, railway construction (CPR completed 1885), and industrialization drove rapid growth 1870–1930.
  - Resource extraction (forestry, mining, oil) and manufacturing (auto sector) fuelled 20th-century prosperity.

Table 7.1 – Effects of Different Growth Rates on Living Standards



After 70 years at 7% growth, income is ≈**8**× higher than at 2% growth.

Why Do Growth Rates Matter?

- Small differences in annual growth rates lead to huge differences in living standards.
- A country that grows too slowly fails to raise living standards:
  - Not just about smartphones — about basic indicators:
    - Infant mortality, life expectancy, access to clean water, health care, education.
- **Example contrast (2023):**
  - Canada: real GDP per capita $\approx$ \$53,000 (2012 \$); infant mortality $\approx$ 4.5/1,000.
  - Sub-Saharan Africa average: $\approx$ \$4,000; infant mortality $\approx$ 50–80/1,000.
- **Canadian concern:** Canada's labour productivity growth of only 1.4%/year (2000–2023) is among the lowest in the G7 — raising long-run living standard concerns.
  - Source: Statistics Canada, Centre for the Study of Living Standards.

Why Do Growth Rates Matter?

- Small differences in annual growth rates lead to **huge** differences in living standards.
- A country that grows too slowly fails to raise living standards:
  - Not just about smartphones — about basic indicators:
    - infant mortality, life expectancy, access to clean water, health care, education.
- **Example contrast (2023):**
  - Canada: real GDP per capita $\approx$ \$53,000 (2012 \$); infant mortality $\approx$ 4.5/1,000.
  - Sub-Saharan Africa average: $\approx$ \$4,000; infant mortality $\approx$ 50–80/1,000.
- **Canadian concern:** Canada's labour productivity growth of only 1.4%/year (2000–2023) is among the lowest in the G7 — raising long-run living standard concerns.
  - Source: Statistics Canada; Centre for the Study of Living Standards.

Why Do Growth Rates Matter?

- Small differences in annual growth rates lead to huge differences in living standards.
- A country that grows too slowly fails to raise living standards:
  - Not just about smartphones — about basic indicators:
    - infant mortality, life expectancy, access to clean water, health care, education.
- **Example contrast (2023):**
  - Canada: real GDP per capita ≈ \$53,000 (2012 \$); infant mortality ≈ 4.5/1,000.
  - Sub-Saharan Africa average: ≈ \$4,000; infant mortality ≈ 50–80/1,000.
- **Canadian concern:** Canada's labour productivity growth of only 1.4%/year (2000–2023) is among the lowest in the G7 — raising long-run living standard concerns.
  - Source: Statistics Canada, Centre for the Study of Living Standards.

Why Do Growth Rates Matter?

- Small differences in annual growth rates lead to huge differences in living standards.
- A country that grows too slowly fails to raise living standards:
  - Not just about smartphones — about basic indicators:
    - infant mortality, life expectancy, access to clean water, health care, education.
- **Example contrast (2023):**
  - Canada: real GDP per capita $\approx$ \$53,000 (2012 \$); infant mortality $\approx$ 4.5/1,000.
  - Sub-Saharan Africa average: $\approx$ \$4,000; infant mortality $\approx$ 50–80/1,000.
- **Canadian concern:** Canada's labour productivity growth of only 1.4%/year (2000–2023) is among the lowest in the G7 — raising long-run living standard concerns.
  - Source: Statistics Canada, Centre for the Study of Living Standards.

Why Do Growth Rates Matter?

- Small differences in annual growth rates lead to huge differences in living standards.
- A country that grows too slowly fails to raise living standards:
  - Not just about smartphones — about basic indicators:
    - infant mortality, life expectancy, access to clean water, health care, education.
- **Example contrast (2023):**
  - Canada: real GDP per capita $\approx$ \$53,000 (2012 \$); infant mortality $\approx$ 4.5/1,000.
  - Sub-Saharan Africa average: $\approx$ \$4,000; infant mortality $\approx$ 50–80/1,000.
- **Canadian concern:** Canada's labour productivity growth of only 1.4%/year (2000–2023) is among the lowest in the G7 — raising long-run living standard concerns.
  - Source: Statistics Canada, Centre for the Study of Living Standards.

Why Do Growth Rates Matter?

- Small differences in annual growth rates lead to huge differences in living standards.
- A country that grows too slowly fails to raise living standards:
  - Not just about smartphones — about basic indicators:
    - infant mortality, life expectancy, access to clean water, health care, education.
- **Example contrast (2023):**
  - Canada: real GDP per capita $\approx$ \$53,000 (2012 \$); infant mortality $\approx$ 4.5/1,000.
  - Sub-Saharan Africa average: $\approx$ \$4,000; infant mortality $\approx$ 50–80/1,000.
- **Canadian concern:** Canada's labour productivity growth of only 1.4%/year (2000–2023) is among the lowest in the G7 — raising long-run living standard concerns.
  - Source: Statistics Canada; Centre for the Study of Living Standards

Why Do Growth Rates Matter?

- Small differences in annual growth rates lead to huge differences in living standards.
- A country that grows too slowly fails to raise living standards:
  - Not just about smartphones — about basic indicators:
    - infant mortality, life expectancy, access to clean water, health care, education.
- **Example contrast (2023):**
  - Canada: real GDP per capita $\approx$ \$53,000 (2012 \$); infant mortality $\approx$ 4.5/1,000.
  - Sub-Saharan Africa average: $\approx$ \$4,000; infant mortality $\approx$ 50–80/1,000.
- **Canadian concern:** Canada's labour productivity growth of only 1.4%/year (2000–2023) is among the lowest in the G7 — raising long-run living standard concerns.
  - Source: Statistics Canada, Centre for the Study of Living Standards.

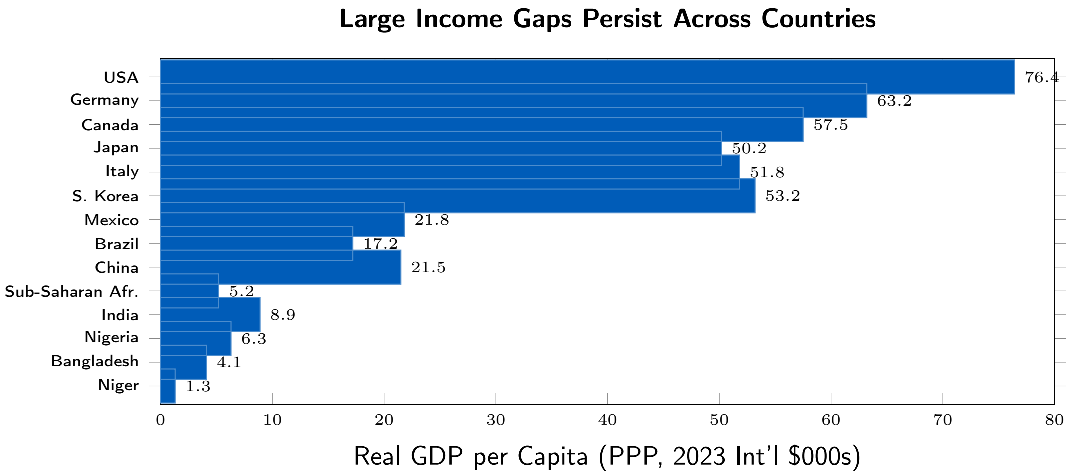
Why Do Growth Rates Matter?

- Small differences in annual growth rates lead to huge differences in living standards.
- A country that grows too slowly fails to raise living standards:
  - Not just about smartphones — about basic indicators:
    - infant mortality, life expectancy, access to clean water, health care, education.
- **Example contrast (2023):**
  - Canada: real GDP per capita ≈ $53,000 (2012 $); infant mortality ≈ 4.5/1,000.
  - Sub-Saharan Africa average: ≈ $4,000; infant mortality ≈ 50–80/1,000.
- **Canadian concern:** Canada's labour productivity growth of only 1.4%/year (2000–2023) is among the lowest in the G7 — raising long-run living standard concerns.
  - Source: Statistics Canada, Centre for the Study of Living Standards.

Why Do Growth Rates Matter?

- Small differences in annual growth rates lead to huge differences in living standards.
- A country that grows too slowly fails to raise living standards:
  - Not just about smartphones — about basic indicators:
    - infant mortality, life expectancy, access to clean water, health care, education.
- **Example contrast (2023):**
  - Canada: real GDP per capita $\approx$ \$53,000 (2012 \$); infant mortality $\approx$ 4.5/1,000.
  - Sub-Saharan Africa average: $\approx$ \$4,000; infant mortality $\approx$ 50–80/1,000.
- **Canadian concern:** Canada's labour productivity growth of only 1.4%/year (2000–2023) is among the lowest in the G7 — raising long-run living standard concerns.
  - Source: Statistics Canada, Centre for the Study of Living Standards.

Figure 7.1 – Real GDP per Capita Around the World (2023)



Source: IMF World Economic Outlook, 2024 (approximate PPP values).

7.2 Learning Objective

Goal

Use the economic growth model to explain why growth rates differ across countries.

- An **economic growth model** explains growth rates in real GDP per capita over the long run.
- Key from Chapter 6: long-run growth in real GDP per person is driven by **labour productivity**.
- Two main factors:
  1. **Quantity of capital available to workers**
  2. **Level of technology**

7.2 Learning Objective

Goal

Use the economic growth model to explain why growth rates differ across countries.

- An **economic growth model** explains growth rates in real GDP per capita over the long run.
- Key from Chapter 6: long-run growth in real GDP per person is driven by **labour productivity**.
- Two main factors:
  1. **Quantity of capital available to workers**
  2. **Level of technology**

7.2 Learning Objective

Goal

Use the economic growth model to explain why growth rates differ across countries.

- An **economic growth model** explains growth rates in real GDP per capita over the long run.
- Key from Chapter 6: long-run growth in real GDP per person is driven by **labour productivity**.
- Two main factors:
  1. **Quantity of capital available to workers**
  2. **Level of technology**

7.2 Learning Objective

Goal

Use the economic growth model to explain why growth rates differ across countries.

- An **economic growth model** explains growth rates in real GDP per capita over the long run.
- Key from Chapter 6: long-run growth in real GDP per person is driven by **labour productivity**.
- Two main factors:
  1. **Quantity of capital available to workers**
  2. **Level of technology**

7.2 Learning Objective

Goal

Use the economic growth model to explain why growth rates differ across countries.

- An **economic growth model** explains growth rates in real GDP per capita over the long run.
- Key from Chapter 6: long-run growth in real GDP per person is driven by **labour productivity**.
- Two main factors:
  1. **Quantity of capital available to workers**
  2. **Level of technology**

Labour Productivity and Its Determinants

- **Labour productivity:**

$$\text{Labour productivity} = \frac{\text{Real GDP}}{\text{Hours worked}}.$$

- Three sources of **technological change**:
  1. **Better machinery and equipment**
     - E.g., AI-assisted design tools, CNC manufacturing, precision agriculture drones.
  2. **Increases in human capital**
     - E.g., University of BC STEM graduates adopting quantum computing.
  3. **Better organization and management**
     - E.g., Amazon's logistics algorithms, Toyota's lean production.

Labour Productivity and Its Determinants

- **Labour productivity:**

$$\text{Labour productivity} = \frac{\text{Real GDP}}{\text{Hours worked}}.$$

- Three sources of **technological change**:
  1. **Better machinery and equipment**
     - E.g., AI-assisted design tools, CNC manufacturing, precision agriculture drones.
  2. **Increases in human capital**
     - E.g., University of BC STEM graduates adopting quantum computing.
  3. **Better organization and management**
     - E.g., Amazon's logistics algorithms, Toyota's lean production.

Labour Productivity and Its Determinants

- **Labour productivity:**

$$\text{Labour productivity} = \frac{\text{Real GDP}}{\text{Hours worked}}.$$

- Three sources of **technological change**:
  1. **Better machinery and equipment**
     - E.g., AI-assisted design tools, CNC manufacturing, precision agriculture drones.
  2. **Increases in human capital**
     - E.g., University of BC STEM graduates adopting quantum computing.
  3. **Better organization and management**
     - E.g., Amazon's logistics algorithms, Toyota's lean production.

Labour Productivity and Its Determinants

- **Labour productivity:**

$$\text{Labour productivity} = \frac{\text{Real GDP}}{\text{Hours worked}}.$$

- Three sources of **technological change**:
  1. **Better machinery and equipment**
     - E.g., AI-assisted design tools, CNC manufacturing, precision agriculture drones.
  2. **Increases in human capital**
     - E.g., University of BC STEM graduates adopting quantum computing.
  3. **Better organization and management**
     - E.g., Amazon's logistics algorithms, Toyota's lean production.

Labour Productivity and Its Determinants

- **Labour productivity:**

$$\text{Labour productivity} = \frac{\text{Real GDP}}{\text{Hours worked}}.$$

- Three sources of **technological change**:
  1. **Better machinery and equipment**
     - E.g., AI-assisted design tools, CNC manufacturing, precision agriculture drones.
  2. **Increases in human capital**
     - E.g., University of BC STEM graduates adopting quantum computing.
  3. **Better organization and management**
     - E.g., Amazon's logistics algorithms, Toyota's lean production.

Labour Productivity and Its Determinants

- **Labour productivity:**

$$\text{Labour productivity} = \frac{\text{Real GDP}}{\text{Hours worked}}.$$

- Three sources of **technological change**:
  1. **Better machinery and equipment**
     - E.g., AI-assisted design tools, CNC manufacturing, precision agriculture drones.
  2. **Increases in human capital**
     - E.g., University of BC STEM graduates adopting quantum computing.
  3. **Better organization and management**
     - E.g., Amazon's logistics algorithms, Toyota's lean production.

Labour Productivity and Its Determinants

- **Labour productivity:**

$$\text{Labour productivity} = \frac{\text{Real GDP}}{\text{Hours worked}}.$$

- Three sources of **technological change**:
  1. **Better machinery and equipment**
     - E.g., AI-assisted design tools, CNC manufacturing, precision agriculture drones.
  2. **Increases in human capital**
     - E.g., University of BC STEM graduates adopting quantum computing.
  3. **Better organization and management**
     - E.g., Amazon's logistics algorithms, Toyota's lean production.

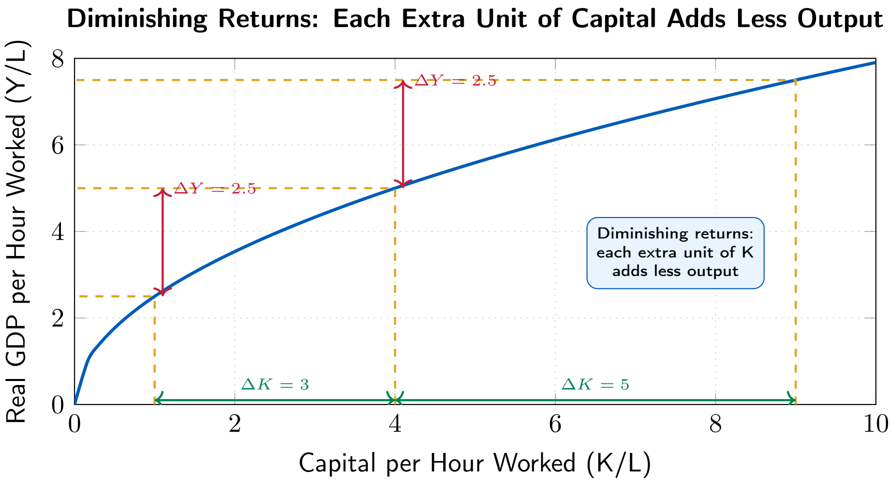
Labour Productivity and Its Determinants

- **Labour productivity:**

$$\text{Labour productivity} = \frac{\text{Real GDP}}{\text{Hours worked}}.$$

- Three sources of **technological change**:
  1. **Better machinery and equipment**
     - E.g., AI-assisted design tools, CNC manufacturing, precision agriculture drones.
  2. **Increases in human capital**
     - E.g., University of BC STEM graduates adopting quantum computing.
  3. **Better organization and management**
     - E.g., Amazon's logistics algorithms, Toyota's lean production.

Figure 7.2 – The Per-Worker Production Function



Diminishing Returns to Capital

- **Diminishing returns:** increasing capital progressively while holding other inputs constant yields **smaller and smaller** increases in output.
- Intuition:
  - Give a forestry worker their first chainsaw: huge productivity gain.
  - Give them a second chainsaw: modest gain (can't use both simultaneously).
  - Give them the tenth identical chainsaw: almost zero extra gain.
- **Implication:**
  - Capital deepening alone cannot sustain high growth forever.
  - Poorer countries (low K/L) gain more from each unit of new capital than rich countries.
  - Rich countries (high K/L) must rely more heavily on **technological change**.

Diminishing Returns to Capital

- **Diminishing returns:** increasing capital progressively while holding other inputs constant yields **smaller and smaller** increases in output.
- Intuition:
  - Give a forestry worker their first chainsaw: huge productivity gain.
  - Give them a second chainsaw: modest gain (can't use both simultaneously).
  - Give them the tenth identical chainsaw: almost zero extra gain.
- **Implication:**
  - Capital deepening alone cannot sustain high growth forever.
  - Poorer countries (low K/L) gain more from each unit of new capital than rich countries.
  - Rich countries (high K/L) must rely more heavily on **technological change**.

Diminishing Returns to Capital

- **Diminishing returns:** increasing capital progressively while holding other inputs constant yields **smaller and smaller** increases in output.
- Intuition:
  - Give a forestry worker their first chainsaw: huge productivity gain.
  - Give them a second chainsaw: modest gain (can't use both simultaneously).
  - Give them the tenth identical chainsaw: almost zero extra gain.
- **Implication:**
  - Capital deepening alone cannot sustain high growth forever.
  - Poorer countries (low K/L) gain more from each unit of new capital than rich countries.
  - Rich countries (high K/L) must rely more heavily on **technological change**.

Diminishing Returns to Capital

- **Diminishing returns:** increasing capital progressively while holding other inputs constant yields **smaller and smaller** increases in output.
- Intuition:
  - Give a forestry worker their first chainsaw: huge productivity gain.
  - Give them a second chainsaw: modest gain (can't use both simultaneously).
  - Give them the tenth identical chainsaw: almost zero extra gain.
- **Implication:**
  - Capital deepening alone cannot sustain high growth forever.
  - Poorer countries (low K/L) gain more from each unit of new capital than rich countries.
  - Rich countries (high K/L) must rely more heavily on **technological change**.

Diminishing Returns to Capital

- **Diminishing returns:** increasing capital progressively while holding other inputs constant yields **smaller and smaller** increases in output.
- Intuition:
  - Give a forestry worker their first chainsaw: huge productivity gain.
  - Give them a second chainsaw: modest gain (can't use both simultaneously).
  - Give them the tenth identical chainsaw: almost zero extra gain.
- **Implication:**
  - Capital deepening alone cannot sustain high growth forever.
  - Poorer countries (low K/L) gain more from each unit of new capital than rich countries.
  - Rich countries (high K/L) must rely more heavily on **technological change**.

Diminishing Returns to Capital

- **Diminishing returns:** increasing capital progressively while holding other inputs constant yields **smaller and smaller** increases in output.
- Intuition:
  - Give a forestry worker their first chainsaw: huge productivity gain.
  - Give them a second chainsaw: modest gain (can't use both simultaneously).
  - Give them the tenth identical chainsaw: almost zero extra gain.
- **Implication:**
  - Capital deepening alone cannot sustain high growth forever.
  - Poorer countries (low K/L) gain more from each unit of new capital than rich countries.
  - Rich countries (high K/L) must rely more heavily on **technological change**.

Diminishing Returns to Capital

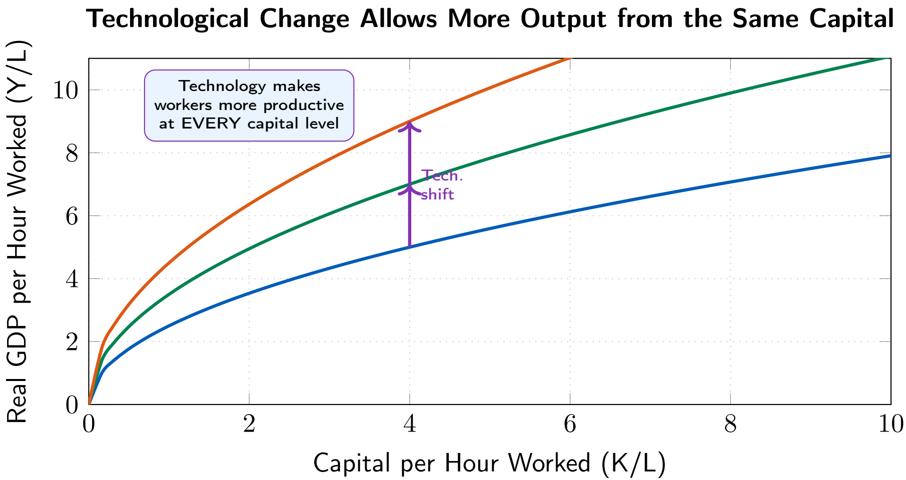
- **Diminishing returns:** increasing capital progressively while holding other inputs constant yields **smaller and smaller** increases in output.
- Intuition:
  - Give a forestry worker their first chainsaw: huge productivity gain.
  - Give them a second chainsaw: modest gain (can't use both simultaneously).
  - Give them the tenth identical chainsaw: almost zero extra gain.
- **Implication:**
  - Capital deepening alone cannot sustain high growth forever.
  - Poorer countries (low K/L) gain more from each unit of new capital than rich countries.
  - Rich countries (high K/L) must rely more heavily on **technological change**.

Diminishing Returns to Capital

- **Diminishing returns:** increasing capital progressively while holding other inputs constant yields **smaller and smaller** increases in output.
- Intuition:
  - Give a forestry worker their first chainsaw: huge productivity gain.
  - Give them a second chainsaw: modest gain (can't use both simultaneously).
  - Give them the tenth identical chainsaw: almost zero extra gain.
- **Implication:**
  - Capital deepening alone cannot sustain high growth forever.
  - Poorer countries (low K/L) gain more from each unit of new capital than rich countries.
  - Rich countries (high K/L) must rely more heavily on **technological change**.

Diminishing Returns to Capital

- **Diminishing returns:** increasing capital progressively while holding other inputs constant yields **smaller and smaller** increases in output.
- Intuition:
  - Give a forestry worker their first chainsaw: huge productivity gain.
  - Give them a second chainsaw: modest gain (can't use both simultaneously).
  - Give them the tenth identical chainsaw: almost zero extra gain.
- **Implication:**
  - Capital deepening alone cannot sustain high growth forever.
  - Poorer countries (low K/L) gain more from each unit of new capital than rich countries.
  - Rich countries (high K/L) must rely more heavily on **technological change**.


Technological Change Allows More Output from the Same Capital
Technology makes workers more productive at EVERY capital level
Tech. shift
Real GDP per Hour Worked (Y/L)
Capital per Hour Worked (K/L)
0
2
4
6
8
10

Capital Deepening vs. Technological Change

Capital Deepening

- Increase in K/L (moving along the production function).
- Faces **diminishing returns**.
- Powerful for poor countries (low K/L).
- Weakens in rich countries.

Technological Change

- Shifts the production function **upward**.
- No inherent diminishing returns.
- Essential for sustained growth in rich countries.
- Driven by R&D, education, entrepreneurship.

- **Canadian example:** Calgary's oil sector deepened physical capital in the 1990s–2000s (oilsands expansion). Waterloo's tech corridor relies on technological change (BlackBerry, Miovision, AI startups).

7.3 Learning Objective

Goal

Explain the role of knowledge capital and government policy in sustaining long-run growth.

- The Solow model (1950s): technological change is exogenous — it "just happens."
- **New growth theory** (Paul Romer, 1990s):
  - Technological change is driven by **economic incentives** — it is endogenous.
  - Firms and researchers respond to profit opportunities.
  - Romer won the Nobel Prize in Economics in 2018 for this contribution.

7.3 Learning Objective

Goal

Explain the role of knowledge capital and government policy in sustaining long-run growth.

- The Solow model (1950s): technological change is exogenous — it "just happens."
- **New growth theory** (Paul Romer, 1990s):
  - Technological change is driven by **economic incentives** — it is endogenous.
  - Firms and researchers respond to profit opportunities.
  - Romer won the Nobel Prize in Economics in 2018 for this contribution.

7.3 Learning Objective

Goal

Explain the role of knowledge capital and government policy in sustaining long-run growth.

- The Solow model (1950s): technological change is exogenous — it "just happens."
- **New growth theory** (Paul Romer, 1990s):
  - Technological change is driven by **economic incentives** — it is endogenous.
  - Firms and researchers respond to profit opportunities.
  - Romer won the Nobel Prize in Economics in 2018 for this contribution.

7.3 Learning Objective

Goal

Explain the role of knowledge capital and government policy in sustaining long-run growth.

- The Solow model (1950s): technological change is exogenous — it "just happens."
- **New growth theory** (Paul Romer, 1990s):
  - Technological change is driven by **economic incentives** — it is endogenous.
  - Firms and researchers respond to profit opportunities.
  - Romer won the Nobel Prize in Economics in 2018 for this contribution.

7.3 Learning Objective

Goal

Explain the role of knowledge capital and government policy in sustaining long-run growth.

- The Solow model (1950s): technological change is exogenous — it "just happens."
- **New growth theory** (Paul Romer, 1990s):
  - Technological change is driven by **economic incentives** — it is endogenous.
  - Firms and researchers respond to profit opportunities.
  - Romer won the Nobel Prize in Economics in 2018 for this contribution.

Knowledge Capital

- **Knowledge capital:** the stock of ideas, scientific knowledge, designs, and blueprints.
  - Generated by R&D, learning-by-doing, spillovers.
- **Key distinction from physical capital:**
  - Physical capital is **rival**: a tractor used by Farmer A cannot simultaneously be used by Farmer B.
  - Knowledge is **nonrival**: the recipe for mRNA vaccines (Moderna) can be used simultaneously by hundreds of manufacturers worldwide.
- **Canadian R&D example:**
  - University of Toronto's CIFAR AI lab developed foundational deep learning (Hinton, LeCun, Bengio) — knowledge that now powers billions of applications globally.
  - This knowledge was **nonrival**: using it in Google's search doesn't prevent Meta from using it.

Knowledge Capital

- **Knowledge capital:** the stock of ideas, scientific knowledge, designs, and blueprints.
  - Generated by R&D, learning-by-doing, spillovers.
- **Key distinction from physical capital:**
  - Physical capital is **rival**: a tractor used by Farmer A cannot simultaneously be used by Farmer B.
  - Knowledge is **nonrival**: the recipe for mRNA vaccines (Moderna) can be used simultaneously by hundreds of manufacturers worldwide.
- **Canadian R&D example:**
  - University of Toronto's CIFAR AI lab developed foundational deep learning (Hinton, LeCun, Bengio) — knowledge that now powers billions of applications globally.
  - This knowledge was **nonrival**: using it in Google's search doesn't prevent Meta from using it.

Knowledge Capital

- **Knowledge capital:** the stock of ideas, scientific knowledge, designs, and blueprints.
    - Generated by R&D, learning-by-doing, spillovers.
- **Key distinction from physical capital:**
    - Physical capital is **rival**: a tractor used by Farmer A cannot simultaneously be used by Farmer B.
    - Knowledge is **nonrival**: the recipe for mRNA vaccines (Moderna) can be used simultaneously by hundreds of manufacturers worldwide.
- **Canadian R&D example:**
    - University of Toronto's CIFAR AI lab developed foundational deep learning (Hinton, LeCun, Bengio) — knowledge that now powers billions of applications globally.
    - This knowledge was **nonrival**: using it in Google's search doesn't prevent Meta from using it.

Knowledge Capital

- **Knowledge capital:** the stock of ideas, scientific knowledge, designs, and blueprints.
  - Generated by R&D, learning-by-doing, spillovers.
- **Key distinction from physical capital:**
  - Physical capital is **rival**: a tractor used by Farmer A cannot simultaneously be used by Farmer B.
  - Knowledge is **nonrival**: the recipe for mRNA vaccines (Moderna) can be used simultaneously by hundreds of manufacturers worldwide.
- **Canadian R&D example:**
  - University of Toronto's CIFAR AI lab developed foundational deep learning (Hinton, LeCun, Bengio) — knowledge that now powers billions of applications globally.
  - This knowledge was **nonrival**: using it in Google's search doesn't prevent Meta from using it.

Knowledge Capital

- **Knowledge capital:** the stock of ideas, scientific knowledge, designs, and blueprints.
  - Generated by R&D, learning-by-doing, spillovers.
- **Key distinction from physical capital:**
  - Physical capital is **rival**: a tractor used by Farmer A cannot simultaneously be used by Farmer B.
  - Knowledge is **nonrival**: the recipe for mRNA vaccines (Moderna) can be used simultaneously by hundreds of manufacturers worldwide.
- **Canadian R&D example:**
  - University of Toronto's CIFAR AI lab developed foundational deep learning (Hinton, LeCun, Bengio) — knowledge that now powers billions of applications globally.
  - This knowledge was **nonrival**: using it in Google's search doesn't prevent Meta from using it.

Knowledge Capital

- **Knowledge capital:** the stock of ideas, scientific knowledge, designs, and blueprints.
  - Generated by R&D, learning-by-doing, spillovers.
- **Key distinction from physical capital:**
  - Physical capital is **rival**: a tractor used by Farmer A cannot simultaneously be used by Farmer B.
  - Knowledge is **nonrival**: the recipe for mRNA vaccines (Moderna) can be used simultaneously by hundreds of manufacturers worldwide.
- **Canadian R&D example:**
  - University of Toronto's CIFAR AI lab developed foundational deep learning (Hinton, LeCun, Bengio) — knowledge that now powers billions of applications globally.
  - This knowledge was **nonrival**: using it in Google's search doesn't prevent Meta from using it.

Knowledge Capital

- **Knowledge capital:** the stock of ideas, scientific knowledge, designs, and blueprints.
  - Generated by R&D, learning-by-doing, spillovers.
- **Key distinction from physical capital:**
  - Physical capital is **rival**: a tractor used by Farmer A cannot simultaneously be used by Farmer B.
  - Knowledge is **nonrival**: the recipe for mRNA vaccines (Moderna) can be used simultaneously by hundreds of manufacturers worldwide.
- **Canadian R&D example:**
  - University of Toronto's CIFAR AI lab developed foundational deep learning (Hinton, LeCun, Bengio) — knowledge that now powers billions of applications globally.
  - This knowledge was **nonrival**: using it in Google's search doesn't prevent Meta from using it.

Knowledge Capital

- **Knowledge capital:** the stock of ideas, scientific knowledge, designs, and blueprints.
  - Generated by R&D, learning-by-doing, spillovers.
- **Key distinction from physical capital:**
  - Physical capital is **rival**: a tractor used by Farmer A cannot simultaneously be used by Farmer B.
  - Knowledge is **nonrival**: the recipe for mRNA vaccines (Moderna) can be used simultaneously by hundreds of manufacturers worldwide.
- **Canadian R&D example:**
  - University of Toronto's CIFAR AI lab developed foundational deep learning (Hinton, LeCun, Bengio) — knowledge that now powers billions of applications globally.
  - This knowledge was **nonrival**: using it in Google's search doesn't prevent Meta from using it.

Figure 7.4 – Physical Capital vs. Knowledge Capital

Physical Capital

- Rival (one user)
- Excludable
- Diminishing returns
- E.g., machines, trucks

Adding more has decreasing impact

Key

Difference

Knowledge Capital

- Nonrival (many users)
- Often nonexcludable
- Increasing returns possible
- E.g., algorithms, vaccines

Can grow without diminishing returns

Because knowledge is nonrival, it can generate **increasing returns to scale** at the economy-wide level, potentially sustaining growth indefinitely — unlike physical capital.

Free-Rider Problem and Underinvestment in R&D

- Because knowledge capital is a **public good**, firms may not capture its full benefits.
- **Free-rider problem:**
  - Firm A invests \$100 million in R&D to develop a new drug.
  - Competitors can study the drug, reverse-engineer it, and produce cheaper generics.
  - Firm A bears the full cost but competitors capture part of the benefit.
- **Result:** Market produces too little R&D relative to the socially optimal amount.
- **Measurement:** Canada's R&D spending = 1.7% of GDP (2023) — below the OECD average of 2.7% (OECD MSTI, 2024). This gap is a major policy concern.

Free-Rider Problem and Underinvestment in R&D

- Because knowledge capital is a **public good**, firms may not capture its full benefits.
- **Free-rider problem:**
  - Firm A invests \$100 million in R&D to develop a new drug.
  - Competitors can study the drug, reverse-engineer it, and produce cheaper generics.
  - Firm A bears the full cost but competitors capture part of the benefit.
- **Result:** Market produces too little R&D relative to the socially optimal amount.
- **Measurement:** Canada's R&D spending = 1.7% of GDP (2023) — below the OECD average of 2.7% (OECD MSTI, 2024). This gap is a major policy concern.

Free-Rider Problem and Underinvestment in R&D

- Because knowledge capital is a **public good**, firms may not capture its full benefits.
- **Free-rider problem:**
  - Firm A invests $100 million in R&D to develop a new drug.
  - Competitors can study the drug, reverse-engineer it, and produce cheaper generics.
  - Firm A bears the full cost but competitors capture part of the benefit.
- **Result:** Market produces too little R&D relative to the socially optimal amount.
- **Measurement:** Canada's R&D spending = 1.7% of GDP (2023) — below the OECD average of 2.7% (OECD MSTI, 2024). This gap is a major policy concern.

Free-Rider Problem and Underinvestment in R&D

- Because knowledge capital is a **public good**, firms may not capture its full benefits.
- **Free-rider problem:**
  - Firm A invests $100 million in R&D to develop a new drug.
  - Competitors can study the drug, reverse-engineer it, and produce cheaper generics.
  - Firm A bears the full cost but competitors capture part of the benefit.
- **Result:** Market produces too little R&D relative to the socially optimal amount.
- **Measurement:** Canada's R&D spending = 1.7% of GDP (2023) — below the OECD average of 2.7% (OECD MSTI, 2024). This gap is a major policy concern.

Free-Rider Problem and Underinvestment in R&D

- Because knowledge capital is a **public good**, firms may not capture its full benefits.
- **Free-rider problem:**
  - Firm A invests $100 million in R&D to develop a new drug.
  - Competitors can study the drug, reverse-engineer it, and produce cheaper generics.
  - Firm A bears the full cost but competitors capture part of the benefit.
- **Result:** Market produces too little R&D relative to the socially optimal amount.
- **Measurement:** Canada's R&D spending = 1.7% of GDP (2023) — below the OECD average of 2.7% (OECD MSTI, 2024). This gap is a major policy concern.

Free-Rider Problem and Underinvestment in R&D

- Because knowledge capital is a **public good**, firms may not capture its full benefits.
- **Free-rider problem:**
  - Firm A invests $100 million in R&D to develop a new drug.
  - Competitors can study the drug, reverse-engineer it, and produce cheaper generics.
  - Firm A bears the full cost but competitors capture part of the benefit.
- **Result:** Market produces too little R&D relative to the socially optimal amount.
- **Measurement:** Canada's R&D spending = 1.7% of GDP (2023) — below the OECD average of 2.7% (OECD MSTI, 2024). This gap is a major policy concern.

Free-Rider Problem and Underinvestment in R&D

- Because knowledge capital is a **public good**, firms may not capture its full benefits.
- **Free-rider problem:**
  - Firm A invests $100 million in R&D to develop a new drug.
  - Competitors can study the drug, reverse-engineer it, and produce cheaper generics.
  - Firm A bears the full cost but competitors capture part of the benefit.
- **Result:** Market produces too little R&D relative to the socially optimal amount.
- **Measurement:** Canada's R&D spending = 1.7% of GDP (2023) — below the OECD average of 2.7% (OECD MSTI, 2024). This gap is a major policy concern.

Government's Role: Three Channels

1. **Protecting intellectual property**
   - **Patents:** 20-year exclusive right to produce an invention.
     - Encourages innovation by guaranteeing temporary monopoly profits.
     - E.g., Moderna's mRNA patent gave it time to recoup COVID-19 vaccine R&D.
   - **Copyrights:** protects creative works (books, software, music).
   - Trade-off: strong IP protection boosts innovation but limits diffusion.
2. **Supporting R&D directly**
   - NRC Canada, NSERC grants, SR&ED tax credit ($3–4B annually).
3. **Subsidizing education**
   - Provincial tuition subsidies, federal student loans, immigration of skilled workers.

Government's Role: Three Channels

1. **Protecting intellectual property**
   - **Patents:** 20-year exclusive right to produce an invention.
     - Encourages innovation by guaranteeing temporary monopoly profits.
     - E.g., Moderna's mRNA patent gave it time to recoup COVID-19 vaccine R&D.
   - **Copyrights:** protects creative works (books, software, music).
   - Trade-off: strong IP protection boosts innovation but limits diffusion.
2. **Supporting R&D directly**
   - NRC Canada, NSERC grants, SR&ED tax credit ($3–4B annually).
3. **Subsidizing education**
   - Provincial tuition subsidies, federal student loans, immigration of skilled workers.

Government's Role: Three Channels

1. **Protecting intellectual property**
   - **Patents:** 20-year exclusive right to produce an invention.
     - Encourages innovation by guaranteeing temporary monopoly profits.
     - E.g., Moderna's mRNA patent gave it time to recoup COVID-19 vaccine R&D.
   - **Copyrights:** protects creative works (books, software, music).
   - Trade-off: strong IP protection boosts innovation but limits diffusion.
2. **Supporting R&D directly**
   - NRC Canada, NSERC grants, SR&ED tax credit ($3–4B annually).
3. **Subsidizing education**
   - Provincial tuition subsidies, federal student loans, immigration of skilled workers.

Government's Role: Three Channels

1. **Protecting intellectual property**
   - **Patents:** 20-year exclusive right to produce an invention.
     - Encourages innovation by guaranteeing temporary monopoly profits.
     - E.g., Moderna's mRNA patent gave it time to recoup COVID-19 vaccine R&D.
   - **Copyrights:** protects creative works (books, software, music).
   - Trade-off: strong IP protection boosts innovation but limits diffusion.
2. **Supporting R&D directly**
   - NRC Canada, NSERC grants, SR&ED tax credit ($3–4B annually).
3. **Subsidizing education**
   - Provincial tuition subsidies, federal student loans, immigration of skilled workers.

Government's Role: Three Channels

1. **Protecting intellectual property**
   - **Patents:** 20-year exclusive right to produce an invention.
     - Encourages innovation by guaranteeing temporary monopoly profits.
     - E.g., Moderna's mRNA patent gave it time to recoup COVID-19 vaccine R&D.
   - **Copyrights:** protects creative works (books, software, music).
   - Trade-off: strong IP protection boosts innovation but limits diffusion.
2. **Supporting R&D directly**
   - NRC Canada, NSERC grants, SR&ED tax credit (\$3–4B annually).
3. **Subsidizing education**
   - Provincial tuition subsidies, federal student loans, immigration of skilled workers.

Government's Role: Three Channels

1. **Protecting intellectual property**
   - **Patents:** 20-year exclusive right to produce an invention.
     - Encourages innovation by guaranteeing temporary monopoly profits.
     - E.g., Moderna's mRNA patent gave it time to recoup COVID-19 vaccine R&D.
   - **Copyrights:** protects creative works (books, software, music).
   - Trade-off: strong IP protection boosts innovation but limits diffusion.
2. Supporting R&D directly
   - NRC Canada, NSERC grants, SR&ED tax credit ($3–4B annually).
3. Subsidizing education
   - Provincial tuition subsidies, federal student loans, immigration of skilled workers.

Government's Role: Three Channels

1. **Protecting intellectual property**
   - **Patents:** 20-year exclusive right to produce an invention.
     - Encourages innovation by guaranteeing temporary monopoly profits.
     - E.g., Moderna's mRNA patent gave it time to recoup COVID-19 vaccine R&D.
   - **Copyrights:** protects creative works (books, software, music).
   - Trade-off: strong IP protection boosts innovation but limits diffusion.
2. **Supporting R&D directly**
   - NRC Canada, NSERC grants, SR&ED tax credit ($3–4B annually).
3. **Subsidizing education**
   - Provincial tuition subsidies, federal student loans, immigration of skilled workers.

Government's Role: Three Channels

1. **Protecting intellectual property**
   - **Patents:** 20-year exclusive right to produce an invention.
     - Encourages innovation by guaranteeing temporary monopoly profits.
     - E.g., Moderna's mRNA patent gave it time to recoup COVID-19 vaccine R&D.
   - **Copyrights:** protects creative works (books, software, music).
   - Trade-off: strong IP protection boosts innovation but limits diffusion.
2. **Supporting R&D directly**
   - NRC Canada, NSERC grants, SR&ED tax credit ($3–4B annually).
3. **Subsidizing education**
   - Provincial tuition subsidies, federal student loans, immigration of skilled workers.

Government's Role: Three Channels

1. **Protecting intellectual property**
   - **Patents:** 20-year exclusive right to produce an invention.
     - Encourages innovation by guaranteeing temporary monopoly profits.
     - E.g., Moderna's mRNA patent gave it time to recoup COVID-19 vaccine R&D.
   - **Copyrights:** protects creative works (books, software, music).
   - Trade-off: strong IP protection boosts innovation but limits diffusion.
2. **Supporting R&D directly**
   - NRC Canada, NSERC grants, SR&ED tax credit ($3–4B annually).
3. **Subsidizing education**
   - Provincial tuition subsidies, federal student loans, immigration of skilled workers.

Government's Role: Three Channels

1. **Protecting intellectual property**
   - **Patents:** 20-year exclusive right to produce an invention.
     - Encourages innovation by guaranteeing temporary monopoly profits.
     - E.g., Moderna's mRNA patent gave it time to recoup COVID-19 vaccine R&D.
   - **Copyrights:** protects creative works (books, software, music).
   - Trade-off: strong IP protection boosts innovation but limits diffusion.
2. **Supporting R&D directly**
   - NRC Canada, NSERC grants, SR&ED tax credit ($3–4B annually).
3. **Subsidizing education**
   - Provincial tuition subsidies, federal student loans, immigration of skilled workers.

7.4 Learning Objective

Goal

Explain economic catch-up and why many poor countries have not experienced rapid growth.

- The growth model predicts **catch-up (convergence)**: poor countries should grow faster.
  - Reason: additional capital is more productive at low K/L levels (diminishing returns).
  - Poor countries can **import** technologies from rich countries cheaply.
- Evidence:
  - Among high-income countries, convergence has occurred — South Korea, Taiwan, Ireland.
  - For the world as a whole, convergence has been weak — many poor countries stagnated.

Goal

Explain economic catch-up and why many poor countries have not experienced rapid growth.

- The growth model predicts **catch-up (convergence)**: poor countries should grow faster.
  - Reason: additional capital is more productive at low K/L levels (diminishing returns).
  - Poor countries can **import** technologies from rich countries cheaply.
- Evidence:
  - Among high-income countries, convergence has occurred — South Korea, Taiwan, Ireland.
  - For the world as a whole, convergence has been weak — many poor countries stagnated.

7.4 Learning Objective

Goal

Explain economic catch-up and why many poor countries have not experienced rapid growth.

- The growth model predicts **catch-up (convergence)**: poor countries should grow faster.
  - Reason: additional capital is more productive at low K/L levels (diminishing returns).
  - Poor countries can **import** technologies from rich countries cheaply.
- Evidence:
  - Among high-income countries, convergence has occurred — South Korea, Taiwan, Ireland.
  - For the world as a whole, convergence has been weak — many poor countries stagnated.

7.4 Learning Objective

Goal

Explain economic catch-up and why many poor countries have not experienced rapid growth.

- The growth model predicts **catch-up (convergence)**: poor countries should grow faster.
  - Reason: additional capital is more productive at low K/L levels (diminishing returns).
  - Poor countries can **import** technologies from rich countries cheaply.
- Evidence:
  - Among high-income countries, convergence has occurred — South Korea, Taiwan, Ireland.
  - For the world as a whole, convergence has been weak — many poor countries stagnated.

7.4 Learning Objective

Goal

Explain economic catch-up and why many poor countries have not experienced rapid growth.

- The growth model predicts **catch-up (convergence)**: poor countries should grow faster.
  - Reason: additional capital is more productive at low K/L levels (diminishing returns).
  - Poor countries can **import** technologies from rich countries cheaply.
- Evidence:
  - Among high-income countries, convergence has occurred — South Korea, Taiwan, Ireland.
  - For the world as a whole, convergence has been weak — many poor countries stagnated.

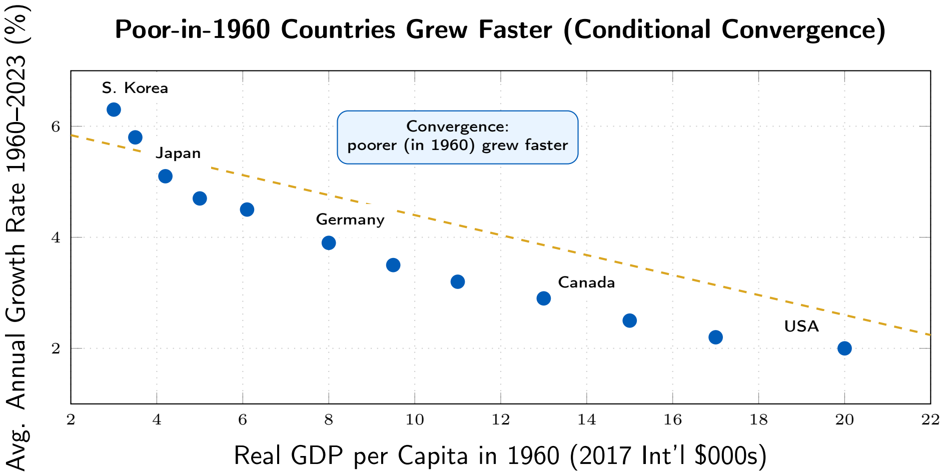
7.4 Learning Objective

Goal

Explain economic catch-up and why many poor countries have not experienced rapid growth.

- The growth model predicts **catch-up (convergence)**: poor countries should grow faster.
  - Reason: additional capital is more productive at low K/L levels (diminishing returns).
  - Poor countries can **import** technologies from rich countries cheaply.
- Evidence:
  - Among high-income countries, convergence has occurred — South Korea, Taiwan, Ireland.
  - For the world as a whole, convergence has been weak — many poor countries stagnated.

Figure 7.5 – Convergence Among High-Income Countries



Sources: Maddison Project; IMF WEO. Each point represents one high-income country.

Why Many Poor Countries Have Not Caught Up

- When we extend the analysis to **all** countries, simple convergence fails:
  - Many low-income countries grew very slowly or experienced **negative** growth.
- Economists highlight four key barriers:

1. Failure of Rule of Law

No property rights, no contract enforcement $\Rightarrow$ entrepreneurs don't invest.

2. Wars & Revolutions

Destroy capital, disrupt trade, create uncertainty $\Rightarrow$ no investment.

3. Poor Education & Health

Low human capital $\Rightarrow$ workers can't adopt new technology.

4. Low Saving & Investment

Vicious cycle: low income $\rightarrow$ low saving $\rightarrow$ low capital $\rightarrow$ low income.

Why Many Poor Countries Have Not Caught Up

- When we extend the analysis to **all** countries, simple convergence fails:
  - Many low-income countries grew very slowly or experienced **negative** growth.
- Economists highlight four key barriers:

1. Failure of Rule of Law

No property rights, no contract enforcement $\Rightarrow$ entrepreneurs don't invest.

2. Wars & Revolutions

Destroy capital, disrupt trade, create uncertainty $\Rightarrow$ no investment.

3. Poor Education & Health

Low human capital $\Rightarrow$ workers can't adopt new technology.

4. Low Saving & Investment

Vicious cycle: low income $\rightarrow$ low saving $\rightarrow$ low capital $\rightarrow$ low income.

Why Many Poor Countries Have Not Caught Up

- When we extend the analysis to **all** countries, simple convergence fails:
  - Many low-income countries grew very slowly or experienced **negative** growth.
- Economists highlight four key barriers:

1. Failure of Rule of Law

No property rights, no contract enforcement $\Rightarrow$ entrepreneurs don't invest.

2. Wars & Revolutions

Destroy capital, disrupt trade, create uncertainty $\Rightarrow$ no investment.

3. Poor Education & Health

Low human capital $\Rightarrow$ workers can't adopt new technology.

4. Low Saving & Investment

Vicious cycle: low income $\rightarrow$ low saving $\rightarrow$ low capital $\rightarrow$ low income.

The Rule of Law: A Prerequisite for Growth

- **Rule of law:** government's ability to enforce laws, especially:
  - protecting **private property rights**,
  - enforcing **contracts**.
- Without strong rule of law:
  - entrepreneurs fear expropriation, corruption, and extortion.
  - They invest **less** in new businesses and technologies.
- **Evidence:** World Bank Doing Business Index and Property Rights Index show strong positive correlations between rule of law and income levels.
- **Canadian context:** Canada consistently ranks among the top 20 globally for rule of law (World Justice Project, 2024) — a key reason it attracts foreign investment.

The Rule of Law: A Prerequisite for Growth

- **Rule of law:** government's ability to enforce laws, especially:
  - protecting **private property rights**,
  - enforcing **contracts**.
- Without strong rule of law:
  - entrepreneurs fear expropriation, corruption, and extortion.
  - They invest **less** in new businesses and technologies.
- **Evidence:** World Bank Doing Business Index and Property Rights Index show strong positive correlations between rule of law and income levels.
- **Canadian context:** Canada consistently ranks among the top 20 globally for rule of law (World Justice Project, 2024) — a key reason it attracts foreign investment.

The Rule of Law: A Prerequisite for Growth

- **Rule of law:** government's ability to enforce laws, especially:
  - protecting **private property rights**,
  - enforcing **contracts**.
- Without strong rule of law:
  - entrepreneurs fear expropriation, corruption, and extortion.
  - They invest **less** in new businesses and technologies.
- **Evidence:** World Bank Doing Business Index and Property Rights Index show strong positive correlations between rule of law and income levels.
- **Canadian context:** Canada consistently ranks among the top 20 globally for rule of law (World Justice Project, 2024) — a key reason it attracts foreign investment.

The Rule of Law: A Prerequisite for Growth

- **Rule of law:** government's ability to enforce laws, especially:
  - protecting **private property rights**,
  - enforcing **contracts**.
- Without strong rule of law:
  - entrepreneurs fear expropriation, corruption, and extortion.
  - They invest **less** in new businesses and technologies.
- **Evidence:** World Bank Doing Business Index and Property Rights Index show strong positive correlations between rule of law and income levels.
- **Canadian context:** Canada consistently ranks among the top 20 globally for rule of law (World Justice Project, 2024) — a key reason it attracts foreign investment.

The Rule of Law: A Prerequisite for Growth

- **Rule of law:** government's ability to enforce laws, especially:
  - protecting **private property rights**,
  - enforcing **contracts**.
- Without strong rule of law:
  - entrepreneurs fear expropriation, corruption, and extortion.
  - They invest **less** in new businesses and technologies.
- **Evidence:** World Bank Doing Business Index and Property Rights Index show strong positive correlations between rule of law and income levels.
- **Canadian context:** Canada consistently ranks among the top 20 globally for rule of law (World Justice Project, 2024) — a key reason it attracts foreign investment.

The Rule of Law: A Prerequisite for Growth

- **Rule of law:** government's ability to enforce laws, especially:
  - protecting **private property rights**,
  - enforcing **contracts**.
- Without strong rule of law:
  - entrepreneurs fear expropriation, corruption, and extortion.
  - They invest **less** in new businesses and technologies.
- **Evidence:** World Bank Doing Business Index and Property Rights Index show strong positive correlations between rule of law and income levels.
- **Canadian context:** Canada consistently ranks among the top 20 globally for rule of law (World Justice Project, 2024) — a key reason it attracts foreign investment.

The Rule of Law: A Prerequisite for Growth

- **Rule of law:** government's ability to enforce laws, especially:
  - protecting **private property rights**,
  - enforcing **contracts**.
- Without strong rule of law:
  - entrepreneurs fear expropriation, corruption, and extortion.
  - They invest **less** in new businesses and technologies.
- **Evidence:** World Bank Doing Business Index and Property Rights Index show strong positive correlations between rule of law and income levels.
- **Canadian context:** Canada consistently ranks among the top 20 globally for rule of law (World Justice Project, 2024) — a key reason it attracts foreign investment.

The Rule of Law: A Prerequisite for Growth

- **Rule of law:** government's ability to enforce laws, especially:
  - protecting **private property rights**,
  - enforcing **contracts**.
- Without strong rule of law:
  - entrepreneurs fear expropriation, corruption, and extortion.
  - They invest **less** in new businesses and technologies.
- **Evidence:** World Bank Doing Business Index and Property Rights Index show strong positive correlations between rule of law and income levels.
- **Canadian context:** Canada consistently ranks among the top 20 globally for rule of law (World Justice Project, 2024) — a key reason it attracts foreign investment.

Why Have Some High-Income Countries Fallen Behind the U.S.?

- Some OECD countries that were close to U.S. productivity in 1990 have **not** fully caught up:
  - Canada's labour productivity (GDP/hour) is approximately 70–75% of U.S. levels.
  - Gap has **widened** since 2000 (Conference Board of Canada, 2024).
- Possible reasons Canada lags:
  - Less competition in key sectors (banking, telecom, airlines) — regulated oligopolies.
  - Lower business R&D investment (1.7% of GDP vs. 3.4% in the U.S.).
  - Higher marginal tax rates on corporate income.
  - Less venture capital availability for startups.
  - Regulatory barriers (interprovincial trade restrictions).

Why Have Some High-Income Countries Fallen Behind the U.S.?

- Some OECD countries that were close to U.S. productivity in 1990 have **not** fully caught up:
  - Canada's labour productivity (GDP/hour) is approximately 70–75% of U.S. levels.
  - Gap has **widened** since 2000 (Conference Board of Canada, 2024).
- Possible reasons Canada lags:
  - Less competition in key sectors (banking, telecom, airlines) — regulated oligopolies.
  - Lower business R&D investment (1.7% of GDP vs. 3.4% in the U.S.).
  - Higher marginal tax rates on corporate income.
  - Less venture capital availability for startups.
  - Regulatory barriers (interprovincial trade restrictions).

Why Have Some High-Income Countries Fallen Behind the U.S.?

- Some OECD countries that were close to U.S. productivity in 1990 have **not** fully caught up:
  - Canada's labour productivity (GDP/hour) is approximately 70–75% of U.S. levels.
  - Gap has **widened** since 2000 (Conference Board of Canada, 2024).
- Possible reasons Canada lags:
  - Less competition in key sectors (banking, telecom, airlines) — regulated oligopolies.
  - Lower business R&D investment (1.7% of GDP vs. 3.4% in the U.S.).
  - Higher marginal tax rates on corporate income.
  - Less venture capital availability for startups.
  - Regulatory barriers (interprovincial trade restrictions).

Why Have Some High-Income Countries Fallen Behind the U.S.?

- Some OECD countries that were close to U.S. productivity in 1990 have **not** fully caught up:
  - Canada's labour productivity (GDP/hour) is approximately 70–75% of U.S. levels.
  - Gap has **widened** since 2000 (Conference Board of Canada, 2024).
- Possible reasons Canada lags:
  - Less competition in key sectors (banking, telecom, airlines) — regulated oligopolies.
  - Lower business R&D investment (1.7% of GDP vs. 3.4% in the U.S.).
  - Higher marginal tax rates on corporate income.
  - Less venture capital availability for startups.
  - Regulatory barriers (interprovincial trade restrictions).

Why Have Some High-Income Countries Fallen Behind the U.S.?

- Some OECD countries that were close to U.S. productivity in 1990 have **not** fully caught up:
  - Canada's labour productivity (GDP/hour) is approximately 70–75% of U.S. levels.
  - Gap has **widened** since 2000 (Conference Board of Canada, 2024).
- Possible reasons Canada lags:
  - Less competition in key sectors (banking, telecom, airlines) — regulated oligopolies.
  - Lower business R&D investment (1.7% of GDP vs. 3.4% in the U.S.).
  - Higher marginal tax rates on corporate income.
  - Less venture capital availability for startups.
  - Regulatory barriers (interprovincial trade restrictions).

Why Have Some High-Income Countries Fallen Behind the U.S.?

- Some OECD countries that were close to U.S. productivity in 1990 have **not** fully caught up:
  - Canada's labour productivity (GDP/hour) is approximately 70–75% of U.S. levels.
  - Gap has **widened** since 2000 (Conference Board of Canada, 2024).
- Possible reasons Canada lags:
  - Less competition in key sectors (banking, telecom, airlines) — regulated oligopolies.
  - Lower business R&D investment (1.7% of GDP vs. 3.4% in the U.S.).
  - Higher marginal tax rates on corporate income.
  - Less venture capital availability for startups.
  - Regulatory barriers (interprovincial trade restrictions).

Why Have Some High-Income Countries Fallen Behind the U.S.?

- Some OECD countries that were close to U.S. productivity in 1990 have **not** fully caught up:
  - Canada's labour productivity (GDP/hour) is approximately 70–75% of U.S. levels.
  - Gap has **widened** since 2000 (Conference Board of Canada, 2024).
- Possible reasons Canada lags:
  - Less competition in key sectors (banking, telecom, airlines) — regulated oligopolies.
  - Lower business R&D investment (1.7% of GDP vs. 3.4% in the U.S.).
  - Higher marginal tax rates on corporate income.
  - Less venture capital availability for startups.
  - Regulatory barriers (interprovincial trade restrictions).

Why Have Some High-Income Countries Fallen Behind the U.S.?

- Some OECD countries that were close to U.S. productivity in 1990 have **not** fully caught up:
  - Canada's labour productivity (GDP/hour) is approximately 70–75% of U.S. levels.
  - Gap has **widened** since 2000 (Conference Board of Canada, 2024).
- Possible reasons Canada lags:
  - Less competition in key sectors (banking, telecom, airlines) — regulated oligopolies.
  - Lower business R&D investment (1.7% of GDP vs. 3.4% in the U.S.).
  - Higher marginal tax rates on corporate income.
  - Less venture capital availability for startups.
  - Regulatory barriers (interprovincial trade restrictions).

Why Have Some High-Income Countries Fallen Behind the U.S.?

- Some OECD countries that were close to U.S. productivity in 1990 have **not** fully caught up:
  - Canada's labour productivity (GDP/hour) is approximately 70–75% of U.S. levels.
  - Gap has **widened** since 2000 (Conference Board of Canada, 2024).
- Possible reasons Canada lags:
  - Less competition in key sectors (banking, telecom, airlines) — regulated oligopolies.
  - Lower business R&D investment (1.7% of GDP vs. 3.4% in the U.S.).
  - Higher marginal tax rates on corporate income.
  - Less venture capital availability for startups.
  - Regulatory barriers (interprovincial trade restrictions).

Summary

- Long-run economic growth is crucial for raising living standards; small differences in growth rates compound into massive income gaps.
- Growth models emphasize:
  - **capital deepening** (more K/L, diminishing returns) and
  - **technological change** (shifts the production function upward).
- **New growth theory** (Romer) places knowledge capital at the centre: it is nonrival, generating potential increasing returns.
- Free-rider problems lead to underinvestment in R&D; governments can help via IP protection, R&D subsidies, and education.
- Convergence holds among high-income countries, but many poor countries are stuck due to:
  - weak rule of law, conflict, poor public services, and low saving.
- Canada's labour productivity gap with the U.S. is a major ongoing policy challenge.

Summary

- Long-run economic growth is crucial for raising living standards; small differences in growth rates compound into massive income gaps.
- Growth models emphasize:
  - **capital deepening** (more K/L, diminishing returns) and
  - **technological change** (shifts the production function upward).
- **New growth theory** (Romer) places knowledge capital at the centre: it is nonrival, generating potential increasing returns.
- Free-rider problems lead to underinvestment in R&D; governments can help via IP protection, R&D subsidies, and education.
- Convergence holds among high-income countries, but many poor countries are stuck due to:
  - weak rule of law, conflict, poor public services, and low saving.
- Canada's labour productivity gap with the U.S. is a major ongoing policy challenge.

Summary

- Long-run economic growth is crucial for raising living standards; small differences in growth rates compound into massive income gaps.
- Growth models emphasize:
  - **capital deepening** (more K/L, diminishing returns) and
  - **technological change** (shifts the production function upward).
- **New growth theory** (Romer) places knowledge capital at the centre: it is nonrival, generating potential increasing returns.
- Free-rider problems lead to underinvestment in R&D; governments can help via IP protection, R&D subsidies, and education.
- Convergence holds among high-income countries, but many poor countries are stuck due to:
  - weak rule of law, conflict, poor public services, and low saving.
- Canada's labour productivity gap with the U.S. is a major ongoing policy challenge.

Summary

- Long-run economic growth is crucial for raising living standards; small differences in growth rates compound into massive income gaps.
- Growth models emphasize:
  - **capital deepening** (more K/L, diminishing returns) and
  - **technological change** (shifts the production function upward).
- **New growth theory** (Romer) places knowledge capital at the centre: it is nonrival, generating potential increasing returns.
- Free-rider problems lead to underinvestment in R&D; governments can help via IP protection, R&D subsidies, and education.
- Convergence holds among high-income countries, but many poor countries are stuck due to:
  - weak rule of law, conflict, poor public services, and low saving.
- Canada's labour productivity gap with the U.S. is a major ongoing policy challenge.

Summary

- Long-run economic growth is crucial for raising living standards; small differences in growth rates compound into massive income gaps.
- Growth models emphasize:
    - **capital deepening** (more K/L, diminishing returns) and
    - **technological change** (shifts the production function upward).
- **New growth theory** (Romer) places knowledge capital at the centre: it is nonrival, generating potential increasing returns.
- Free-rider problems lead to underinvestment in R&D; governments can help via IP protection, R&D subsidies, and education.
- Convergence holds among high-income countries, but many poor countries are stuck due to:
    - weak rule of law, conflict, poor public services, and low saving.
- Canada's labour productivity gap with the U.S. is a major ongoing policy challenge.

Summary

- Long-run economic growth is crucial for raising living standards; small differences in growth rates compound into massive income gaps.
- Growth models emphasize:
  - **capital deepening** (more K/L, diminishing returns) and
  - **technological change** (shifts the production function upward).
- **New growth theory** (Romer) places knowledge capital at the centre: it is nonrival, generating potential increasing returns.
- Free-rider problems lead to underinvestment in R&D; governments can help via IP protection, R&D subsidies, and education.
- Convergence holds among high-income countries, but many poor countries are stuck due to:
  - weak rule of law, conflict, poor public services, and low saving.
- Canada's labour productivity gap with the U.S. is a major ongoing policy challenge.

Summary

- Long-run economic growth is crucial for raising living standards; small differences in growth rates compound into massive income gaps.
- Growth models emphasize:
  - **capital deepening** (more K/L, diminishing returns) and
  - **technological change** (shifts the production function upward).
- **New growth theory** (Romer) places knowledge capital at the centre: it is nonrival, generating potential increasing returns.
- Free-rider problems lead to underinvestment in R&D; governments can help via IP protection, R&D subsidies, and education.
- Convergence holds among high-income countries, but many poor countries are stuck due to:
  - weak rule of law, conflict, poor public services, and low saving.
- Canada's labour productivity gap with the U.S. is a major ongoing policy challenge.

Summary

- Long-run economic growth is crucial for raising living standards; small differences in growth rates compound into massive income gaps.
- Growth models emphasize:
  - **capital deepening** (more K/L, diminishing returns) and
  - **technological change** (shifts the production function upward).
- **New growth theory** (Romer) places knowledge capital at the centre: it is nonrival, generating potential increasing returns.
- Free-rider problems lead to underinvestment in R&D; governments can help via IP protection, R&D subsidies, and education.
- Convergence holds among high-income countries, but many poor countries are stuck due to:
  - weak rule of law, conflict, poor public services, and low saving.
- Canada's labour productivity gap with the U.S. is a major ongoing policy challenge.

Summary

- Long-run economic growth is crucial for raising living standards; small differences in growth rates compound into massive income gaps.
- Growth models emphasize:
  - **capital deepening** (more K/L, diminishing returns) and
  - **technological change** (shifts the production function upward).
- **New growth theory** (Romer) places knowledge capital at the centre: it is nonrival, generating potential increasing returns.
- Free-rider problems lead to underinvestment in R&D; governments can help via IP protection, R&D subsidies, and education.
- Convergence holds among high-income countries, but many poor countries are stuck due to:
  - weak rule of law, conflict, poor public services, and low saving.
- Canada's labour productivity gap with the U.S. is a major ongoing policy challenge.

Thank you!

Questions?